SGA - Structuration – Cours 5

Thomas Chedru, Yann Moysan

Plan du cours de Structuration 2026

- Cours 5 : 22Jan26 - Yann Moysan – Sous-jacents « classiques » et gestion
  - 8h15-11h30
- Cours 6 : 29Jan26- Yann Moysan – Sous-jacents « innovants » et gestion
  - 8h15-11h30
- Cours 7 : 05Feb26 - Yann Moysan – Hybrides et Commodities
  - 13h45-17h00
- EVAL : 10Feb26 17h15-20h15 (TBC)

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1. Dividendes proportionnels

2. Volatilité stochastique

3. Corrélation stochastique

4. Compo/Quanto

Conclusion

Dividendes proportionnels (1/4) : Intuitions & leçons du passé

- Dualité de vision du dividende

Actif de l'économie réelle? ⟷ Un marché d'expert?

- Gestion problématique de la pose de dividende en période de crise

effondrement du marché action

↓

déformation de la pose en book

↓

apparition d'une forte exposition aux dividendes

↓

effondrement des dividendes (contexte économico-politique)

Les modèles au service de l'innovation

Dividendes proportionnels (2/4) : Mode de calcul

Index Dividends aim to replicate the sum of the dividends issued by the members of an index with an ex-date occurring during a calendar year

Illustration : Euro STOXX 50

Equity Index

$$\sum_{1}^{50} \text{Nb of Shares x } \textbf{Shares price} \text{ / Index Divisor} \rightarrow 3\,000 \text{ EUR}$$

Same methodology

Dividends index

$$\sum_{1}^{50} \text{Nb of Shares x } \textbf{DPS*} \text{ / Index Divisor} \rightarrow 115 \text{ EUR}$$

**Dividend per Share*

Source: BNP Paribas EDS. For illustrative purpose only.

Dividendes proportionnels (3/4) : Retour au fondamental

BNP PARIBAS
CORPORATE & INVESTMENT BANKING

BNPParibas Equity Derivatives Strategy

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BID/ASK	Source Bloomberg
Currency	Euro
Bloomberg	refers to aggregated Bloomberg dividend estimates.
	BNP Paribas dividend marked at Concensus level (Source Bloomberg)

This report has been published in accordance with BNP Paribas conflict of interest management. Disclosures relating to BNP Paribas potential conflicts of interest are available at https://eqd.bnpparibas.com/research/compliance.asp

Bloomberg Code	Name	Sector	Eurex Product	Div Frequency	Dividend per share 2015 Bear	2015 Base	2015 Bull	2015 Bloomberg	2015 BID	2015 ASK	2016 Bear	2016 Base	2016 Bull	2016 Bloomberg	2016 BID	2016 ASK
AI FP Equity	AIR LIQUIDE SA	Basic Materials	EUR	Annual	2.30	2.40	2.60	2.55	2.45	2.60	2.29	2.50	2.80	2.70	2.35	2.80
BAS GY Equity	BASF SE	Basic Materials	B2AS	Annual	2.70	2.80	3.00	2.80	2.53	2.91	2.70	2.90	3.10	3.00	2.42	2.78
BAYN GY Equity	BAYER AG-REG	Basic Materials	B2AY	Annual	2.10	2.30	2.40	2.25	2.12	2.44	2.33	2.50	2.70	2.50	2.23	2.57
DPW GY Equity	DEUTSCHE POST AG-REG	Industrials	D2PW	Annual	0.80	0.85	0.95	0.85	0.75	0.98	0.86	0.95	1.00	0.95	0.75	0.97

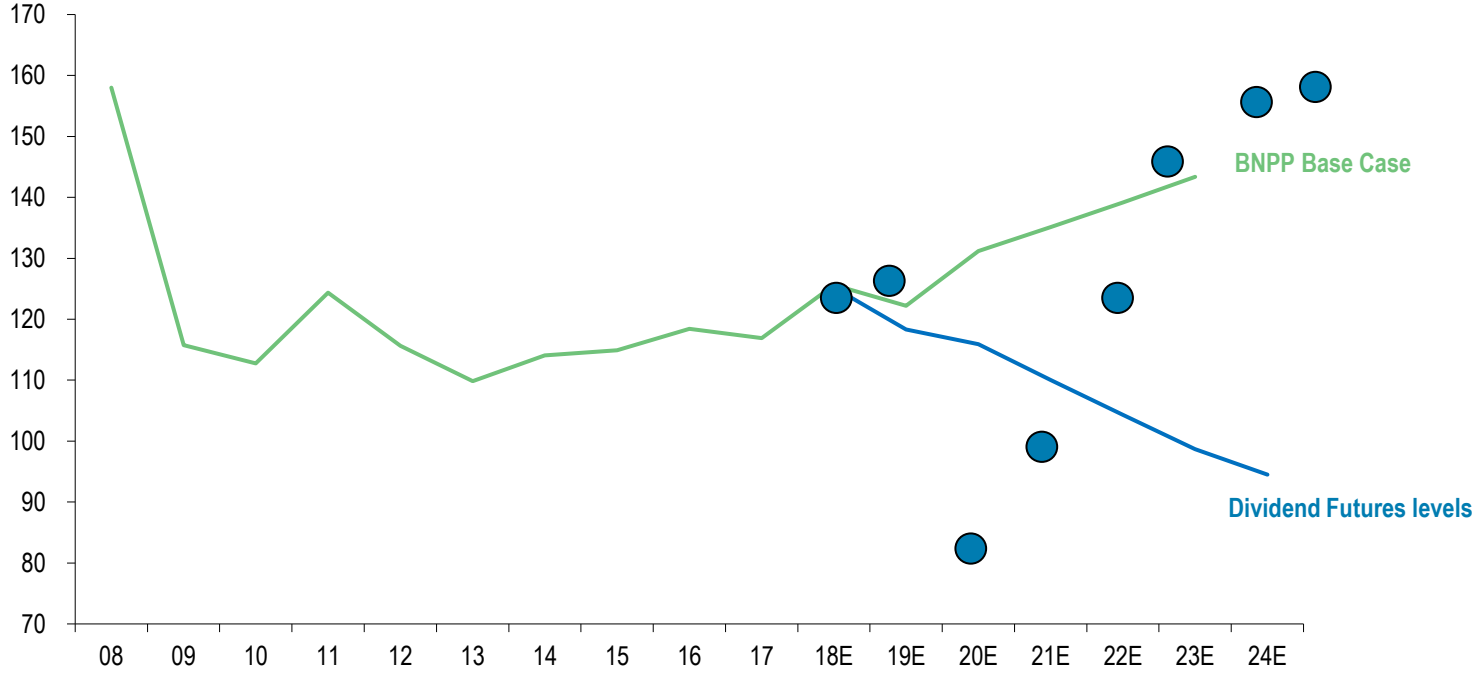
Updated bottom up estimates

	Bear	Base	Bull	Bloomberg
2015	94.5	110.9	119.9	111.3
2016	93.5	117.2	130.4	122.0
2017	97.8	124.0	138.6	132.8

Exemples récents: Actions Bancaires

Les modèles au service de l'innovation

Dividendes proportionnels (4/4) : Ecart Impli vs Estimé



Dividendes proportionnels : Modélisation (1/2)

- Équations de diffusion

$$Div(t) = (1 - \beta(t))Div_{ABS} + \beta(t).Yield.Fwd(t)$$

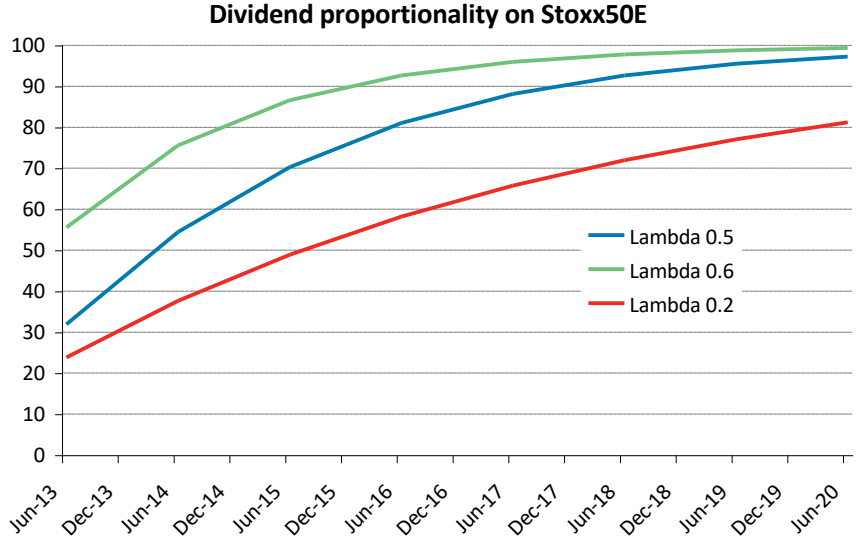
$$\beta(t) = 1 - \exp(-\lambda t)$$

- Distribution réservée

Spot en Hausse -> Dividende en Hausse -> Forward en Baisse
Spot en Baisse -> Dividende en Baisse -> Forward en Hausse

Moins de Vol, besoin d'un refit

- Calibration du lambda



Proportionalité ↗ Delta ↘

Fwd start options!

Les modèles au service de l'innovation

Dividendes proportionnels : Modélisation pour l'Athena (2/2)

Trade details

Strategy code: CE | Legal cpty CRDS: | Marketing client:

Product details

Wrapper: Note | Delivery: Cash | EU Directive: Out Of Scope

Underlying

Underlying:
Basket name: Reload Import

Name	BBG	Spot	Ccy	Tier	Weight
STOXX50E	SX5E	3285.33	EUR	1	

Data Information | Strike per underlying

Early redemption

Autocall trigger (%): 100
Redemption frequency: 1Y
Exit rate (%): 8
Exit rate type: Memory

Airbag details

Airbag coupon: No

Structure details

Issuer: BNP PARIBAS ARI
Detachment rules: None
Detachment date shift: -4
Public offering country: None
Nominal: +3,000,000
Currency: EUR
Currency mode: Quanto

Fixed Coupon

Fixed Coupon: No

Final option

Option type: Put
Option strike: 100
Use downside gearing: No
Downside gearing (%): -100
Barrier: Yes
Barrier monitoring: At Maturity
Knock-in level (%): 80

Delivery details

Settlement in other currency: No

Dates and schedule

Periods
Roll date computation: From Strike date
Pivot date: Observation Date

Delays
Tenor: 5Y
Redemption delay: 10BD

Dates
Strike date: Thu 12 Jan 2017
Issue date: Thu 26 Jan 2017
Valuation date: Wed 12 Jan 2022
Redemption date: Wed 26 Jan 2022

PRICE | **SOLVE**

QUOTED-INDICATIVE PRICE 01/12/2017 17:05

This price is not valid for trading.
Legal counterparty CRDS is mandatory for trading.

Fair Price: 93.68

	AUTO	MANUAL
CC:	NA	6.32
CC @ Risk:	(0.19)	
Dist. fees:	0.00	
Client Price:	NA	100.00

9:41

PRICE
TRADE | RFQ | Missed
Payoff Summary | Get Quote URL

DOCUMENTS

Termsheet 01/12/2017

NOTES

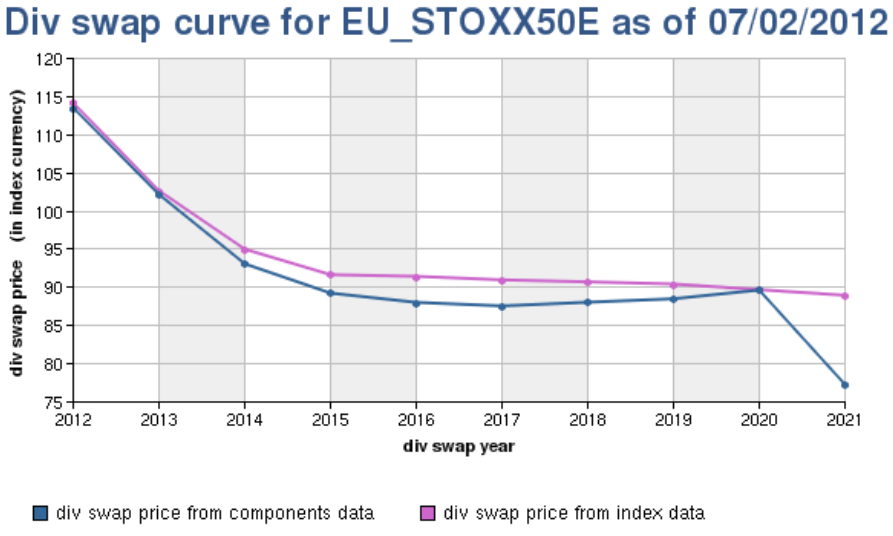
You may enter any comment here

Periods

Observation date	Payment date	Is trigger observed	Trigger level	Exit rate (per period)
Fri 12 Jan 2018	Fri 26 Jan 2018	Yes	100	8
Mon 14 Jan 2019	Mon 28 Jan 2019	Yes	100	16
Mon 13 Jan 2020	Mon 27 Jan 2020	Yes	100	24
Tue 12 Jan 2021	Tue 26 Jan 2021	Yes	100	32
Wed 12 Jan 2022	Wed 26 Jan 2022	Yes	100	40

Dividendes proportionnels : Modélisation, limites et ouverture (3/3)

- Homogénéité du lambda et mismatch Indice vs composants



- Vers un modèle stochastique « SDS »

$$\frac{dDiv(t,T)}{Div(t,T)} = \sigma e^{-\lambda(T-t)} dW_t + \sigma_t^S (1 - e^{-\lambda(T-t)}) dW_t^S$$

- Un candidat idéal pour proposer du Risk-Recycling?

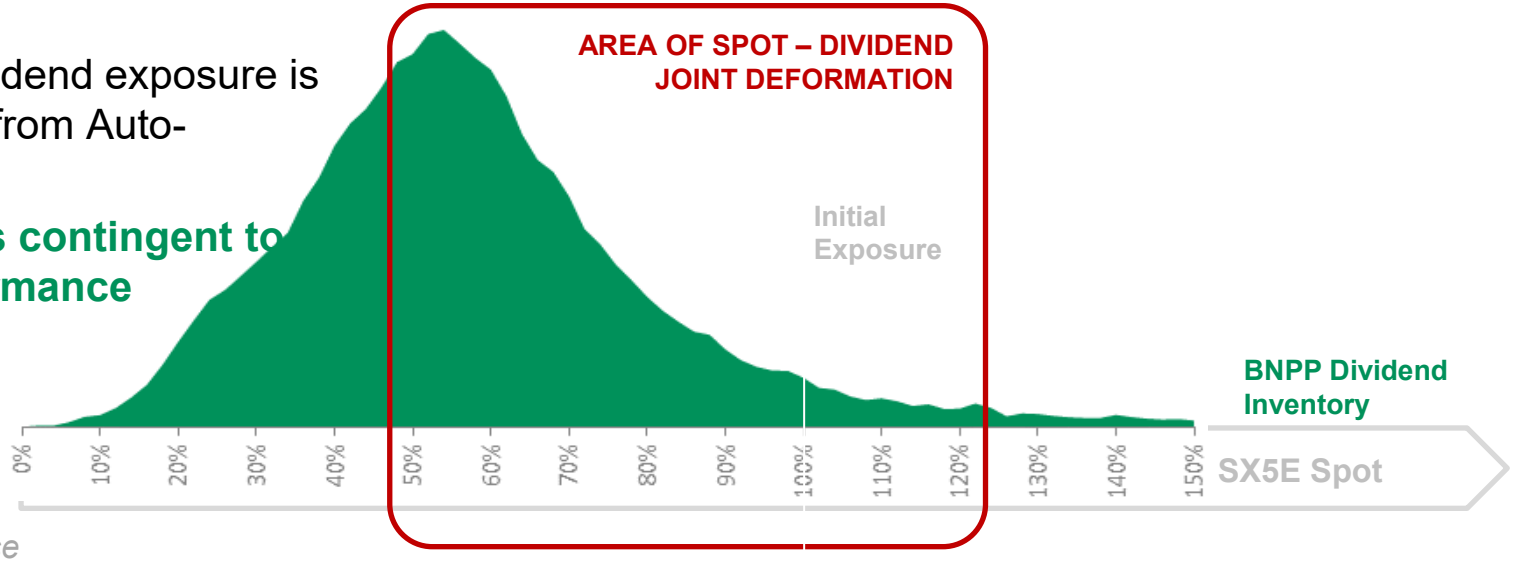
Les modèles au service de l'innovation

Dividendes - Recyclage du risque (1/3) : les origines

A DYNAMIC INVENTORY

BNPP SX5E Dividend exposure is mainly inherited from Auto-Callables

Our Inventory is contingent to the SX5E performance

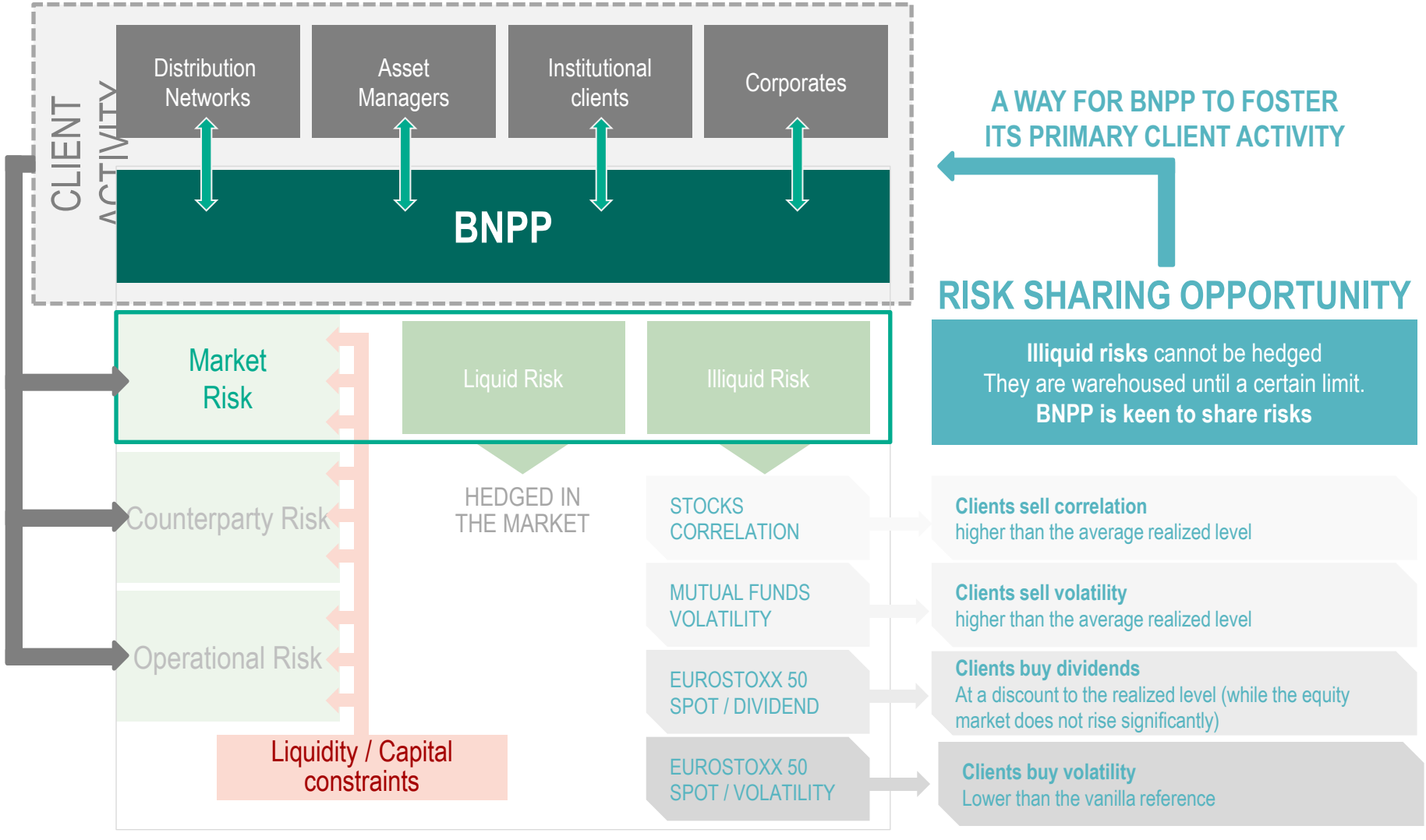


For illustrative purpose

AUTOCALL MECHANISM	**Long SX5E** if SX5E final < 50	**Early redemption** if SX5E annual > 100
BNPP DIVIDEND EXPOSURE	Proportional to SX5E On the downside	Nil On the upside
INVENTORY CONSEQUENCE	Dividend inventory growths when SX5E decreases	Dividend inventory decreases when SX5E increases

Les modèles au service de l'innovation

Dividendes - Recyclage du risque (2/3) : les acteurs ART



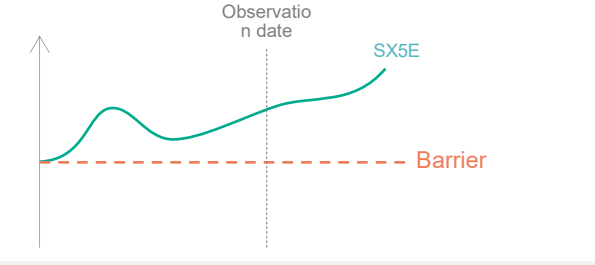
Dividendes - Recyclage du risque (3/3) : Exemple de Transaction ART

U&O FORWARD ON DIVIDEND – TRADE SPECIFICATIONS

LEVERAGED DIVIDEND EXPOSURE AT DISCOUNT

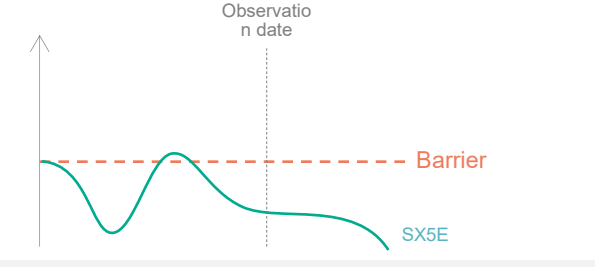
MECHANISM

A product linked to the performance of the SX5E index and the SX5E index dividend future



After the observation date, the SX5E is above the barrier
➔ **The certificate is early redeemed and client receives 100%+Coupon**

Early Payout = 100% + Coupon



After the observation date, the SX5E is below the barrier
➔ **Client is long DEDZ3 performance, with a cap at maturity, with a discount**

Payout = 100% + Min [Cap ; Leverage x ($DEDZ3_{FINAL}$ / $DEDZ3_{INITIAL}$ – Discount)

CHARACTERISTICS

- Format : Certificate
- Maturity : Dec23
- Underlyings: SX5E & DEDZ3
- Observation: After 1Y
- CCY: EUR

Maturity	Dec-23
Coupon	xx%
Barrier	110%
Leverage	2x
Cap	50%
Discount	97%
DEDZ3 (spot)	98.7

As of 8th of May 2018

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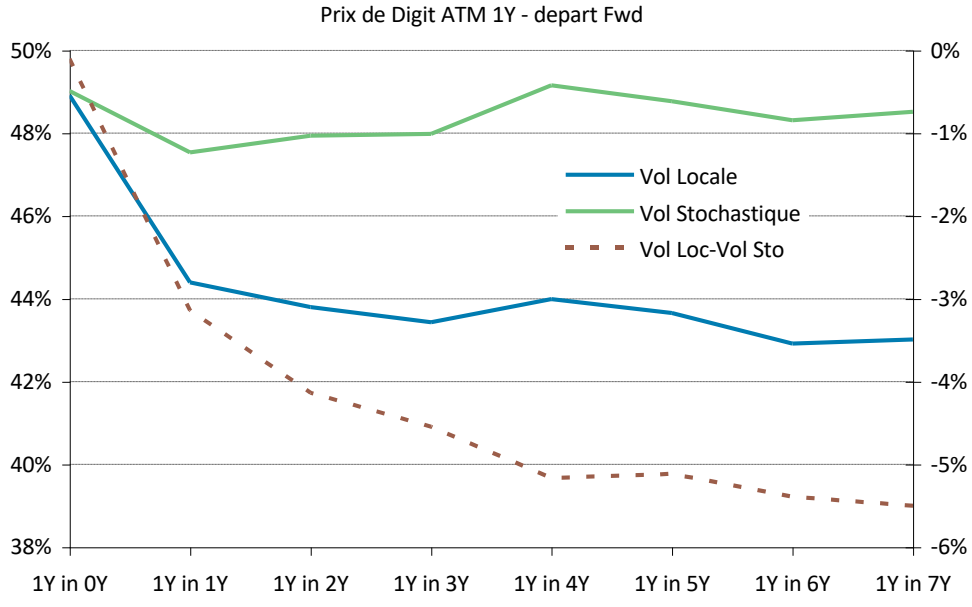
Modèles à Vol Stochastique (1/4) : Intuitions & leçon du passé

- Diffusion des modèles à vol locale

$$\frac{dS_t}{S_t} = \mu dt + \sigma_{loc}(t, S_t) dW_t^S$$

Prix en départ Spot ✓
Prix en départ Forward ✗



- Des produits qui ont fait des dégâts…
  - Cliquets ➡ Smile forward

    Payoff : somme des perf mensuelles localement capée, globalement floorée

  - Twisters & Napoléon ➡ Vovol

    Payoff : Coupon – somme des (ou pire) perf mensuelles, globalement floorée

Modèles à Vol Stochastique (2/4) : Modélisation

- Equations de diffusion

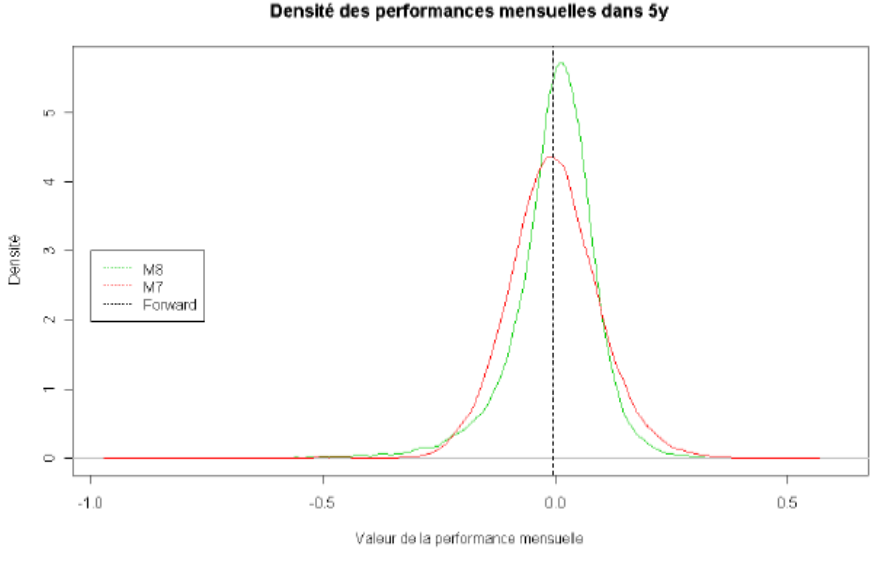
$$\frac{dS_t}{S_t} = \mu dt + \sigma_t dW_t^S \qquad \frac{dV_t(T)}{V_t} = \frac{\omega}{(T-t)^\alpha} dW_t^V$$

vovol

$$\sigma_t = \sqrt{V_{Floor} + V_{(t)}} \qquad \rho = < dW^S, dW^V > \qquad \rho = (1-x)\rho^h - x$$

Correlation Spot-Vol, shiftée

- Distribution des rendements et implications



Distrib in fine vol locale

=

Distrib in fine vol stochastique

⚠

Distrib en cours de vie vol locale

≠

Distrib en cours de vie vol stochastique

Modèles à Vol Stochastique (3/4) : Application et limite

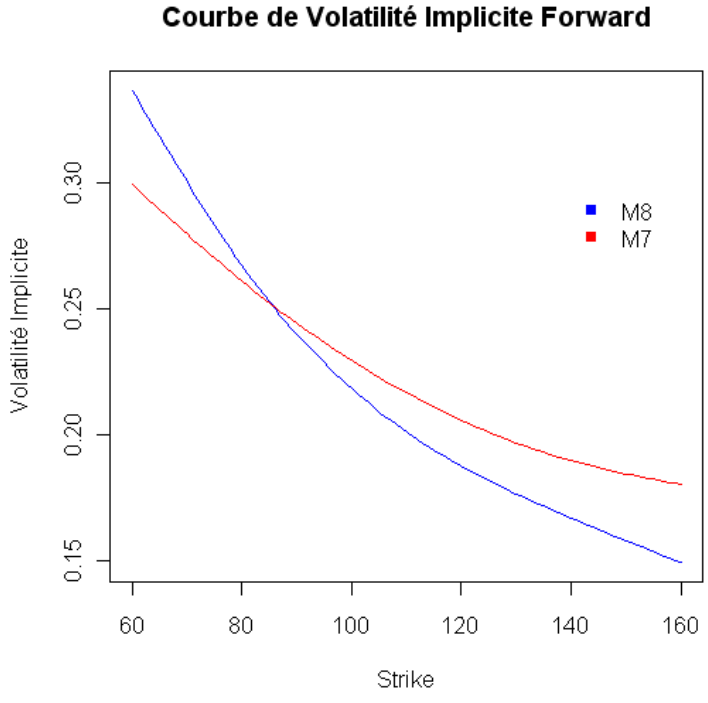
Exemple de produits impactés

- Call sécurisant la performance (aka « Lookback Max »)

  Réduction à une simple digitale Up&In
  Produit moins cher en vol stochastique
  -> compétitif !

- Put à barrière activante (aka « D&I U&O »)

  Les chemins baissiers sont plus volatils
  Produit moins cher en vol stochastique
  -> non compétitif !



Limites du modèles

- Introduction de sauts dans la diffusion
- Perturbation des vanilles : « regriding »
- Gestion en book
- Compétitivité: qui a raison?

Modèles à Vol Stochastique (4/4) : Application et limite

Call lookback yearly SX5E floored at 10% with DO in fine 50% - PDI 50% in fine

Maturité 10Y
Format: Certificat
Maturité: 10 ans
Devise: EUR
Indice: Euro Stoxx 50

Payoff:

Chaque année on enregistre la performance de l'indice depuis le lancement.
le niveau de l'indice est observé à échéance:

A maturité
- Si l'indice est supérieur à 50% de son niveau initial, le produit rembourse 100% du capital investi + 100% de la meilleure performance positive enregistrée, autrement dit : le client sort au meilleur point de sortie enregistré au long de la vie du produit. Floor à l'échéance = 10%

- Sinon le placement est remboursé à la valeur finale de l'indice

Impact du modèle de vol stochastique : 2.20%

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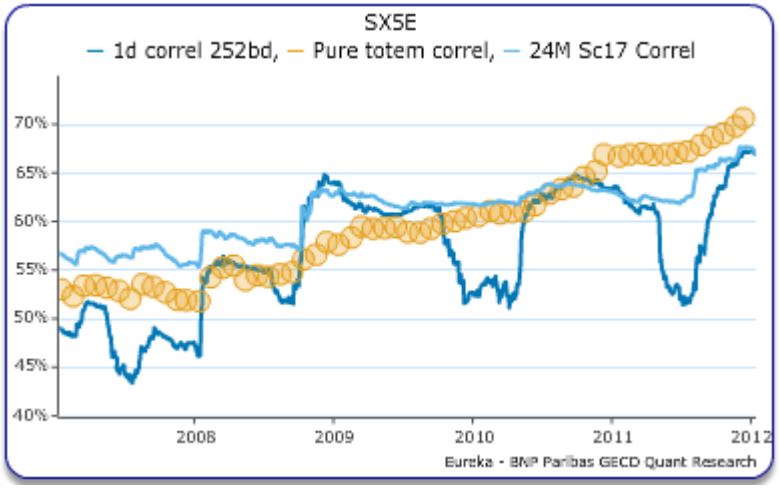
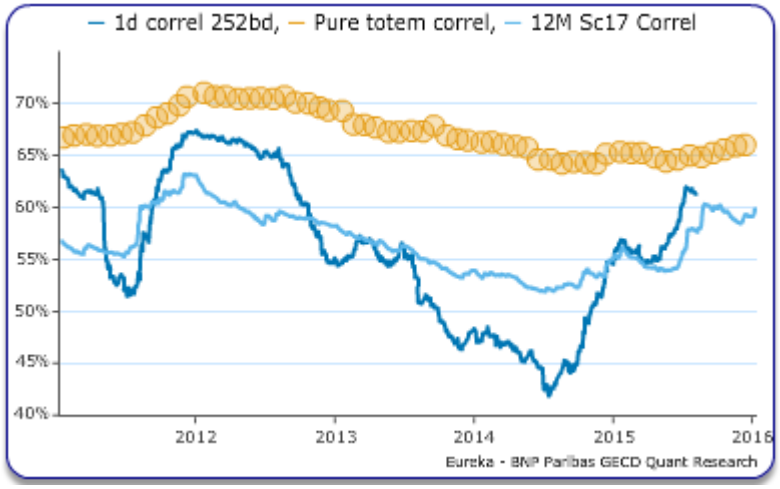
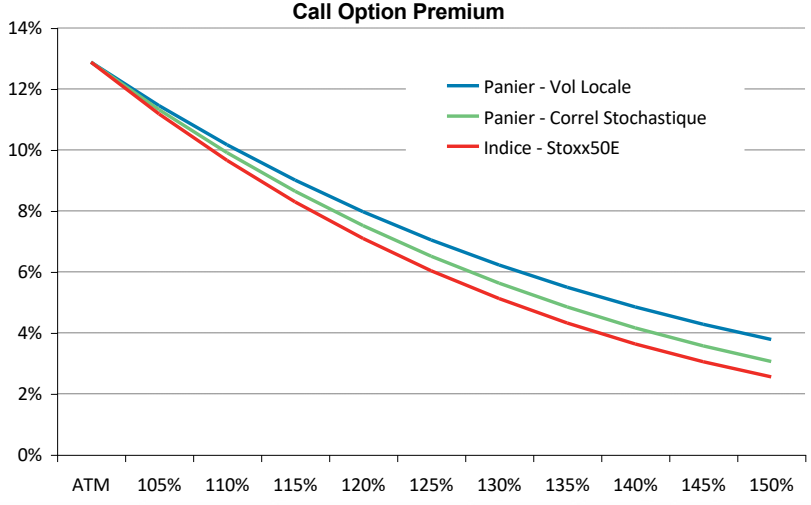
Corrélation Stochastique (1/3) : Intuitions et leçons du passé

- Lien pas si évident entre corrélation et spot → Spot n'est pas l'unique driver





- Mismatch entre Indice et Panier des composants



→ Manque de smile sur les paniers

Corrélation Stochastique (2/3) : Modélisation

- Diffusion : un modèle 1-facteur

$$dW_i = \rho_{iI} dW_I + \varepsilon_i dW_i^{\varepsilon} \quad \Rightarrow \quad dW_i = \frac{\alpha_t \rho_{iI} dW_I + \varepsilon_i dW_i^{\varepsilon}}{\sqrt{\alpha_t^2 \rho_{iI}^2 + \varepsilon_i{}^2}}$$

- Implication pour la corrélation instantanée

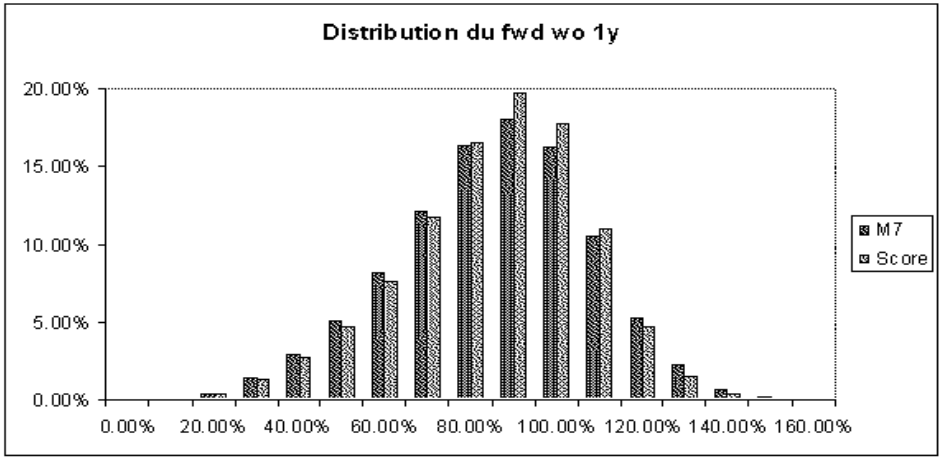
$$\rho_{ij}(t) = < dW_i, dW_j > (t) = \frac{\rho_{ij} + \rho_{iI}\rho_{jI}(f_t^2 - 1)}{\sqrt{1 + \rho_{iI}^2(f_t^2 - 1)}\sqrt{1 + \rho_{jI}^2(f_t^2 - 1)}}$$

Valeur de f_t	Valeur de $\rho_{ij}(t)$
0	Corrélation proche de zéro
1	Corrélation historique ρ_{ij}
$+\infty$	Corrélation de 1

- Distribution réservée

Marché up -> Correl down -> Forward WOF down
Marché down -> Correl up -> Forward WOF up

⇒ Moins de Vol



- Une Matrice modifiée

Correlation Stochastique (3/3) : Applications et Limites du modèle

Produits classiques multi sous-jacents

- Call « Worst-Of »

Option Cash Settlement Amount

On the Cash Settlement Payment Date, the Option Seller will pay to the Option Buyer the Option Cash Settlement Amount calculated by the Calculation Agent as follows:

$$\mathbf{ENA} \times \max\left(0\%, \frac{\mathbf{WO\ Share}_{\text{Final}} - \mathbf{WO\ Share}_{\text{Initial}}}{\mathbf{WO\ Share}_{\text{Initial}}}\right)$$

Where

WO Share is the Share with the worst performance from the **Strike Date** to the **Expiration Date**, defined as:

$$\min_{i=1}^{3}\left(\frac{\mathbf{Share}^{i}_{\text{Final}}}{\mathbf{Share}^{i}_{\text{Initial}}}\right)$$

- Stellar

$$Coupon_t = Max\left[GlobalFloor; \frac{1}{N}\sum_{i=1}^{N} Min\left(LocalCap; \frac{S_{i,t}}{S_{i,Initial}}\right)\right]$$

Limites

- Perturbation -> besoin impact
- Signification financière du facteur
- Modélisation réduite en pratique

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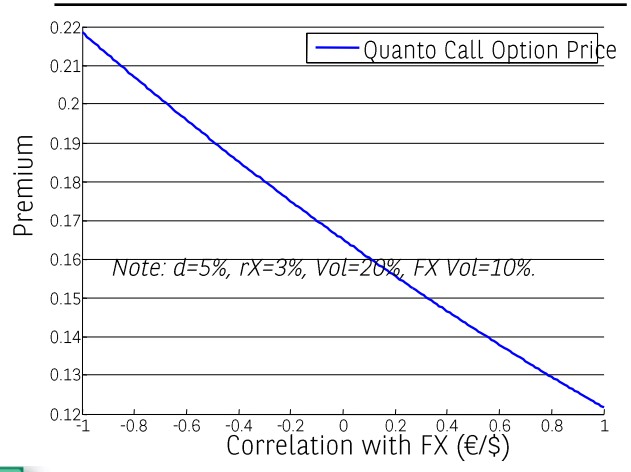
Compo / Quanto (1/3) : Quanto

- quanto is a type of derivative in which the underlying is denominated in one currency (e.g. $), but the instrument itself is settled in another currency (e.g. €) at some fixed rate. Such products are attractive for people who wish to have exposure to a foreign asset, but without the corresponding exchange rate risk.
  - Payoff formula of a quanto call option = $\left[Q_0 \times \max\{0, S(T)_\$ - K_\$\}\right]_{EUR}$
  - Example: S_0 = K = 1$, Q_0 = 0.75, suppose that at T (5y), S(T) = 2$, then at T:

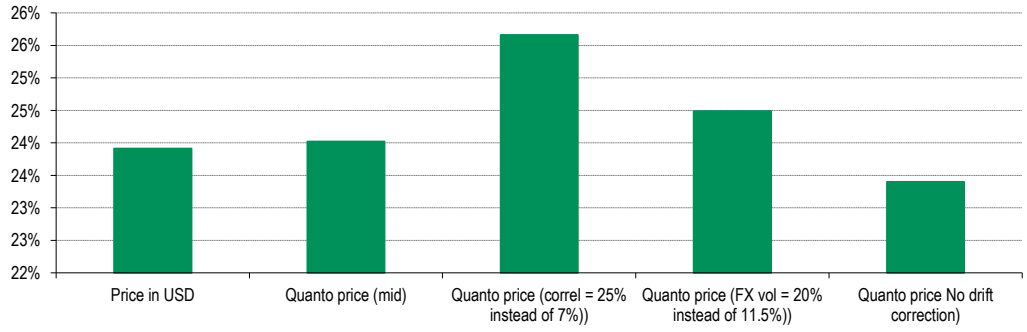
    Payoff = $\left[0.75 \times \max\{0, 2\$ - 1\$\}\right] = 0.75(EUR)$
  - The more correlated FX (€/$) & Equities ($) are, the less the value of a quanto call

    Intuition: When Equity Price ($) grows ➔ FX (USDEUR) increases because of strong correlation, i.e. $ become more expensive compared with € ➔ less valuable the quanto since it pays €.
    (*Similarly, the more correlated FX(EURUSD) & Equities ($) are, the greater the value of a quanto call*)

Quanto Call Price vs. Correlation with FX(USDEUR)



$$dS(t) = (r^X - d - \rho_{S,FX}\sigma_{FX}\nu(t))S(t)dt + \nu(t)S(t)dW_S^{\mathbf{Q}^Y}(t),$$



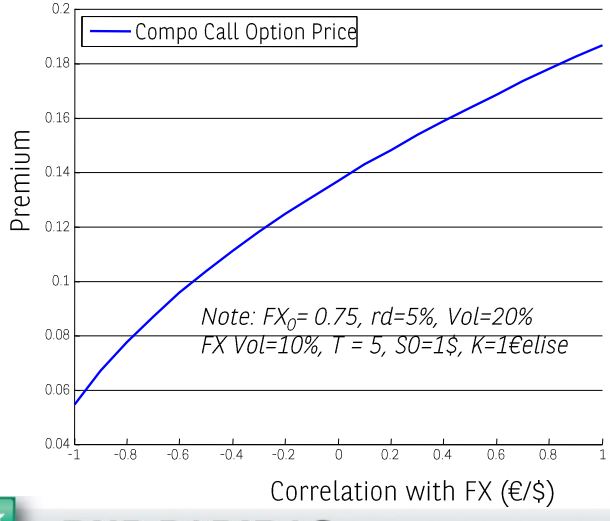
Compo / Quanto (2/3) : Compo

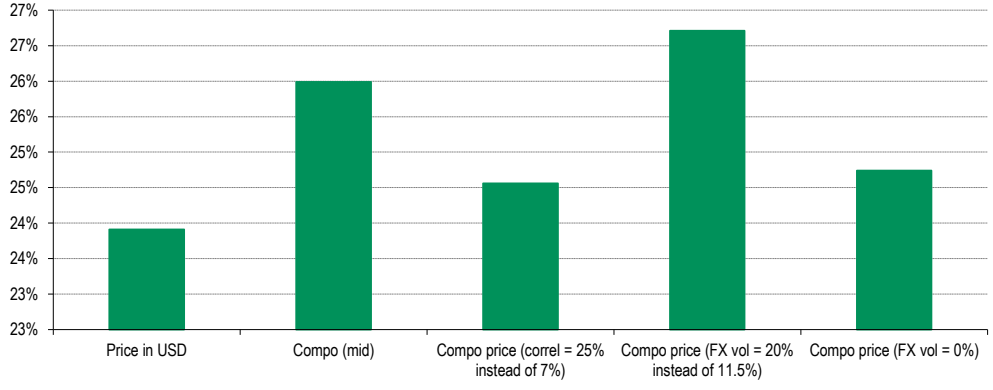
- Option payoff = $MAX\left[FX(T)_{USDEUR} \times S(T)_{USD} - K_{EUR}, 0\right]$
- **Intuition**: we can think SPX(t) and FX(t) (€/$) as a whole as if SPX were traded in euros:
  - At each moment SPXcompo(t) = SPX(t) * USD_EUR(t)
  - SPXcompo is consider as a "basic" underlying in EUR
  - Volatility can be written as:

$$\sigma_{SPXcompo} = \sqrt{\sigma_{SPX}^2 + \sigma_{USDEUR}^2 + 2 \times \sigma_{SPX} \times \sigma_{USDEUR} \times Correl(SPX, USDEUR)}$$

  - Thus, the higher the correlation between Equities and FX(USDEUR), the higher the total volatility ➔ the higher the compo price

Compo Call Price vs. Correlation with FX(USDEUR)



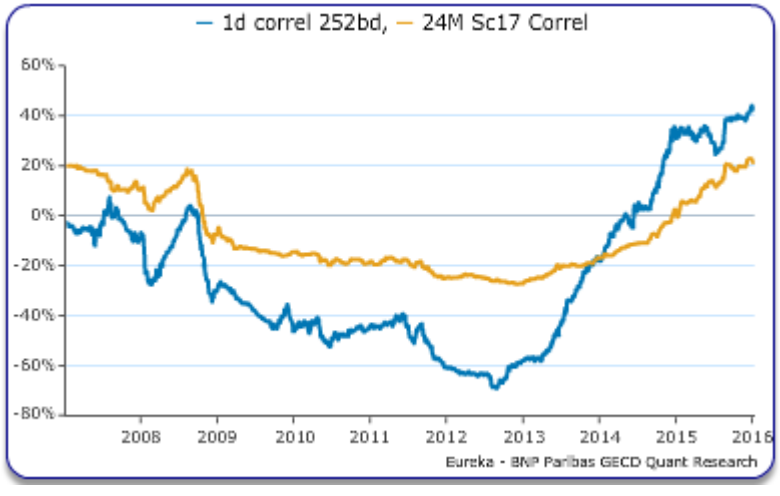


Les modèles au service de l'innovation

Compo / Quanto (3/3) : Modélisation

- Le marking classique n'est plus satisfaisant
  - Au prix sur le compo ou le compo?
  - Quid des politiques monétaires?
- Mise en place d'un modèle original
  - Quels observables? Stabilité?
  - Comment propager?
  - Comment partager?

Correlation SX5E vs USDEUR



Conclusion 1/2

Maturité du business et complexification des modèles….

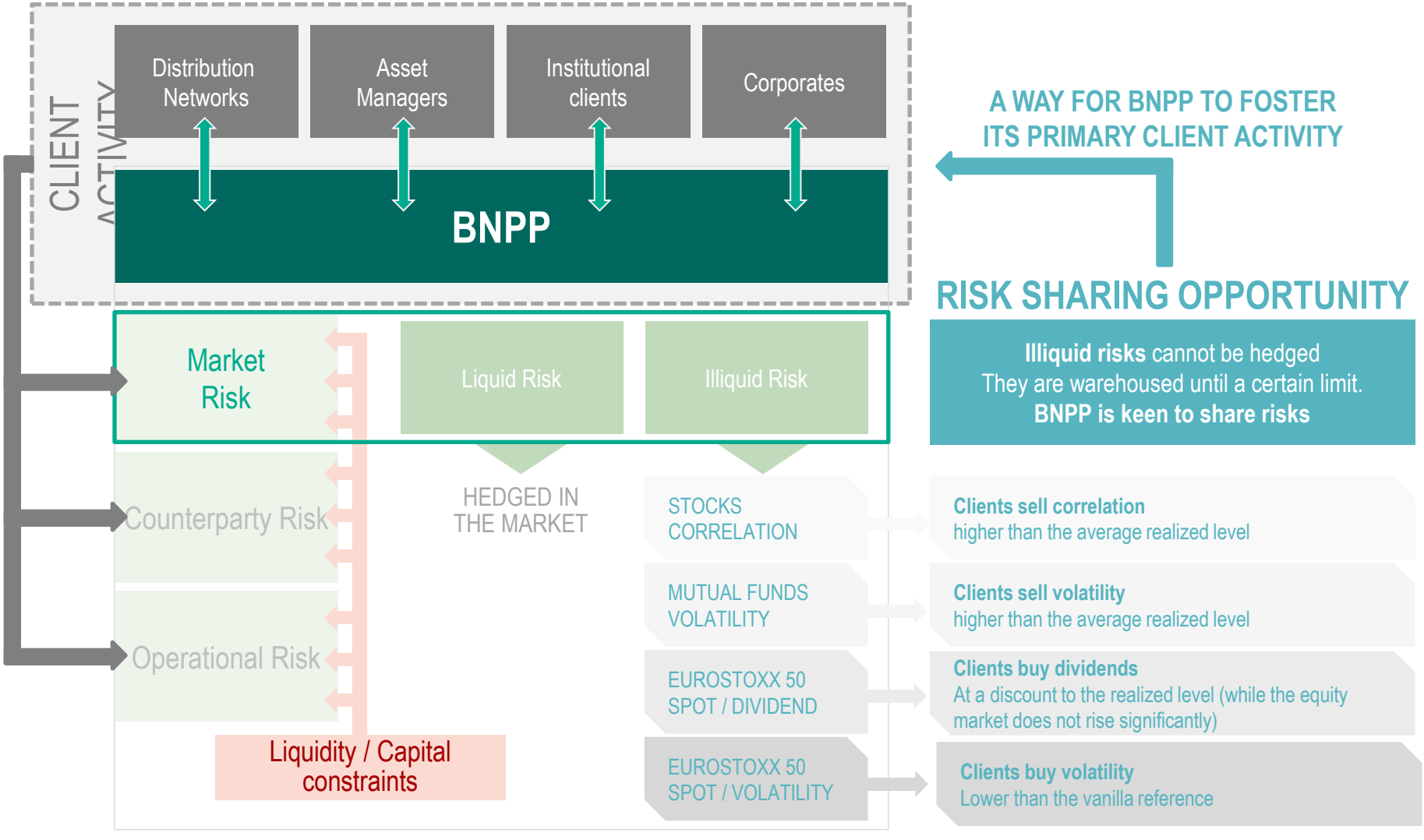
- Compétitivité
- Gestion des produits
- Industrialisation

…pour réussir, le Structureur doit se démarquer par

- Les classes d'actifs proposées
- Des « enveloppes » adaptées
- L'accompagnement des clients
- Le recyclage des risques

Les modèles au service de l'innovation

Conclusion 2/2



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